

FRANK (JAKE) HANG

Incoming Bachelor of Commerce graduate (WAM 74.3) majoring in Economics and Finance, with hands-on experience in portfolio construction, quantitative equity valuation, and econometric modelling. Proficient in R, Python, and Excel, with independent project work spanning Modern Portfolio Theory-based asset allocation, comparable company valuation, and CAPM/factor-based risk analysis across equity markets.

EDUCATION

BACHELOR OF COMMERCE IN ECONOMICS/FINANCE

University of Melbourne

WAM: 74.3

Relevant Coursework: Investments, Corporate Valuation, Econometrics 2, Economic Analysis and Policy, Macroeconomics

expected Nov 2026

Melbourne, VIC

INVESTMENT & QUANTITATIVE PROJECTS

github.com/jahangk/redesigned-lamp

Portfolio Construction & Asset Allocation (R, Python)

- Designed and backtested a Modern Portfolio Theory framework across a 231-asset alternative investment universe, constructing 6 portfolio strategies including rolling- and expanding-window Markowitz tangency portfolios in R
- Built a cap-weighted benchmark index and weekly VWAP aggregation pipeline, integrating a custom Python scraper with historical pricing data to extend time-series coverage through July 2026
- Resolved data continuity artifacts at a multi-source data join, improving the integrity of a multi-year return series used for portfolio backtesting and risk-adjusted performance comparison

Equity Market & Risk Analysis

- Quantified equity market risk exposure by estimating rolling 24-month CAPM and OLS betas for NDQ (NASDAQ 100) and XNT (S&P/ASX 200), identifying macroeconomic drivers behind correlation breakdown between US and Australian equity indices
- Evaluated NAV and index-tracking performance to assess ETF replication efficiency and cross-market diversification benefits, presenting findings and stress-tested claims to a non-technical audience
- Assessed Tesla's market valuation at a ~40% premium to sector median by benchmarking PE multiples against 5 comparable firms, evaluating divergence across growth, margins, and risk profile

Applied Econometrics

- Established a causal link between gender and educational attainment in Tanzania using IV and 2SLS estimation on a 3,500+ observation World Bank dataset, identifying and validating 3 instruments to address OLS endogeneity bias and weak-instrument concerns

EMPLOYMENT HISTORY

1-ON-1 MATHS TUTOR

Private

2022 – 2025

Melbourne, VIC

- Elevated academic performance for 10+ secondary students by an average of 2 letter grades per semester, by designing individualised lesson plans targeting knowledge gaps across algebra, geometry, and statistics
- Sustained a private client base across 3 years through consistent re-engagement, demonstrating reliability, client relationship management, and the ability to build long-term trust
- Adapted quantitative explanations to each student's learning profile, reducing concept-to-application time and improving exam performance across multiple subject areas

TEAM MEMBER

McDonald's

2024 – 2025

Melbourne, VIC

- Served 200+ customers per shift with speed and accuracy, by coordinating order workflows and maintaining real-time communication across a team of 8+ during peak service periods
- Maintained full food safety and hygiene compliance across 12 months of service, by consistently applying quality control procedures in line with operational standards
- Reduced service delays during peak periods by prioritising and sequencing tasks in real time, demonstrating composure and efficiency under high-pressure, time-critical conditions

SKILLS

R, Python, Excel, Portfolio Theory & Asset Allocation, Equity & Comparable Company Valuation, Econometric & Statistical Analysis, Financial Modelling, Policy Analysis

REFERENCES

Available upon request. Academic referee (Econometrics 2 Tutor): University of Melbourne (jiayi.guo.8@unimelb.edu.au)